

Pardeep Kumar

FULL-STACK SOFTWARE ENGINEER | NEXT.JS | TYPESCRIPT | REACT | POSTGRESQL

Amsterdam, the Netherlands | +31 6 1311 9813 | pardeep.iitb@gmail.com | [Portfolio](#)
[LinkedIn](#) | [GitHub](#) | Languages: English, Hindi, Dutch (A2)

PROFILE

Software engineer with more than ten years of experience building applications, data-integration frameworks, simulation platforms, and performance-critical backend components. I independently design, implement, and deploy web products using TypeScript, Next.js, React, PostgreSQL, Python, Rust, real-time services, authentication, automation, and cloud platforms.

My engineering background brings strong architecture, performance, testing, interoperability, and problem-solving habits to product development. I am comfortable carrying an idea from requirements and data modelling through responsive UI, server logic, deployment, observability, and iteration.

FULL-STACK STRENGTHS

Frontend Next.js, React, TypeScript, JavaScript, HTML, CSS, responsive UI, and accessible interaction.	Backend and data Node.js, Python, Rust, PostgreSQL, Drizzle ORM, SQL, REST APIs, server actions, and validation.
Identity and realtime Authentication, permissions, owner-scoped data, Appwrite, transactions, reconnect handling, and persistent sessions.	Deployment Vercel, Cloudflare, Docker, GitHub Actions, environment configuration, migrations, and scheduled publishing.
Architecture Configuration-driven systems, modular monoliths, component registries, data pipelines, and API interoperability.	Broader programming C++, C#, Java, Julia, F#, CUDA, MPI, concurrency, profiling, and performance optimisation.

SELECTED WEB PRODUCTS

WorkAtlas

2026

[Next.js](#) | [React](#) | [TypeScript](#) | [PostgreSQL](#) | [Drizzle ORM](#) | [Live application](#)

- Designed a multi-user workspace for research, projects, publications, experiments, tasks, and long-term ideas.
- Implemented authentication, owner-scoped data, project and Kanban workflows, comments, reminders, documentation, and portable exports.
- Created a configuration-driven content system and deployed it with Vercel and Neon PostgreSQL.

The Republic

2026

[Rust](#) | [Axum](#) | [Tokio](#) | [Phaser 3](#) | [PWA](#) | [Live application](#)

- Built a civic strategy simulation with persistent sessions, idempotent actions, and validated scenario packs.
- Developed a responsive browser client and installable PWA with mobile-ready interaction.

Nazar India

2026

[Python](#) | [Static publishing](#) | [AI-assisted pipeline](#) | [Cloudflare](#) | [Live application](#)

- Built a source-transparent news platform with RSS collection, validation, search, filtering, archives, and structured metadata.
- Automated four-hour collection and publishing workflows with GitHub Actions and Cloudflare.

Chess Duel

2026

[Next.js](#) | [TypeScript](#) | [Appwrite](#) | [Realtime](#) | [Live application](#)

- Implemented server-authoritative chess rules, transaction-safe realtime synchronisation, reconnect handling, and persistent records.
- Supported instant guest play, verified accounts, responsive board orientation, and complete draw conditions.

Pardeep Kumar

SOFTWARE EXPERIENCE SUPPORTING FULL-STACK DELIVERY

PROFESSIONAL EXPERIENCE

PhD Researcher in Scientific Computing

Aug 2022-Aug 14, 2026

Centrum Wiskunde & Informatica (CWI), Amsterdam

- Built maintainable Julia and Python applications, data-generation pipelines, validation workflows, and reproducible computational experiments.
- Delivered an approximately 30x optimisation speedup and developed automated neural-model training and validation workflows.

Software Engineer

Feb-Aug 2022

ASM International, Almere

- Delivered Java simulator features and refactored legacy components for modularity, testability, and release velocity.
- Worked across software and equipment-domain teams to translate operational workflows into application behaviour.

Software Engineer

Jun 2019-Dec 2021

Shell, Amsterdam | Individual contributor

- Designed concurrent integrations between real-time databases and process and pipeline applications.
- Improved throughput by more than 15x and built C#/C++ interoperability, visualisation, and GPU-accelerated components.

Software Engineer

Feb-Apr 2019

Aakraya Research, Bangalore | Team of 2

- Built Python applications for model generation, profit-and-loss calculation, and post-trade analytics.
- Improved C++ order-book and market-data components.

Software Engineer II

May-Nov 2018

KLA-Tencor, Chennai | Team of 2

- Built reusable C# and F# server infrastructure, shared libraries, simulation features, tests, and Erlang fault tolerance.

Software Engineer

Mar 2016-Apr 2018

Altair, Bangalore | Team of 6

- Developed shared-memory infrastructure, engineering UI features, C++/Python integrations, COM APIs, and testing frameworks.

Research and Development Engineer

Jul 2014-Mar 2016

Fluidyn, Bangalore | Team of 4

- Built C++/C# simulation applications, MPI components, reusable numerical libraries, and parallel processing workflows.

ENGINEERING IMPACT

10+ YEARS

Cross-industry software engineering experience

>15X

Higher throughput in a concurrent data-integration workflow

4 PRODUCTS

Independently designed, built, and publicly deployed

EDUCATION

PhD in Mechanical Engineering

2022-2026

TU Delft | Research conducted at CWI Amsterdam

Thesis submitted; defence planned for January 2027.

MSc in Aerospace Engineering

2012-2014

Indian Institute of Technology Bombay, India

Numerical simulation of electromagnetic propagation and scattering.